

EE698G: Circuit Design for Frequency and Phase Synthesis

Project-1: DLL Report

Submitted By:

Samarth Dilip Walse

Roll No: 251040092

Roll No.	Frequency Range
251040092	450 MHz – 500 MHz

1. Voltage-Controlled Delay Line (VCDL) Schematic

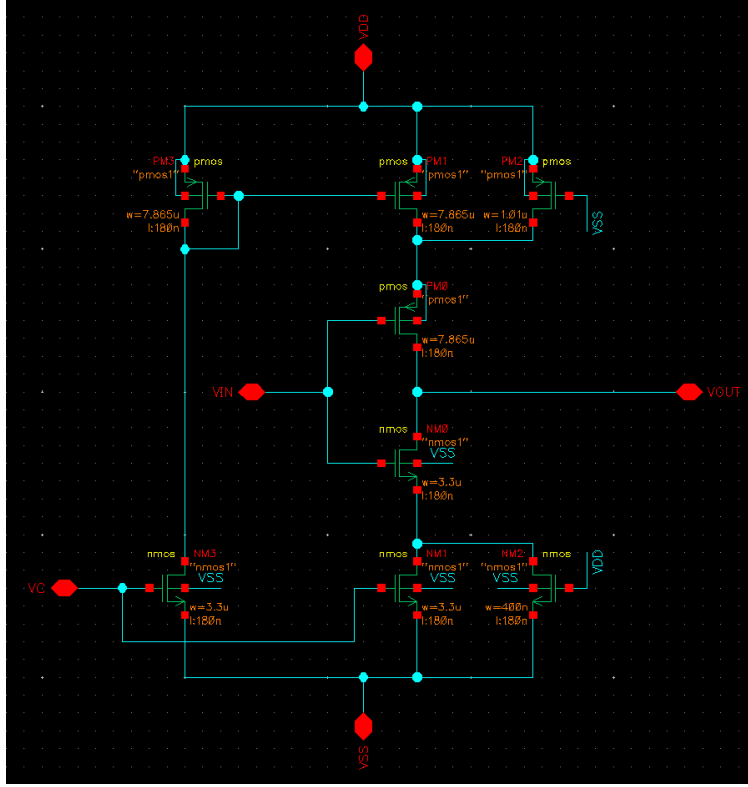


Figure 1: Transistor-level schematic of a single Voltage-Controlled Delay Unit (VCDU), specifying the NMOS and PMOS device sizing.

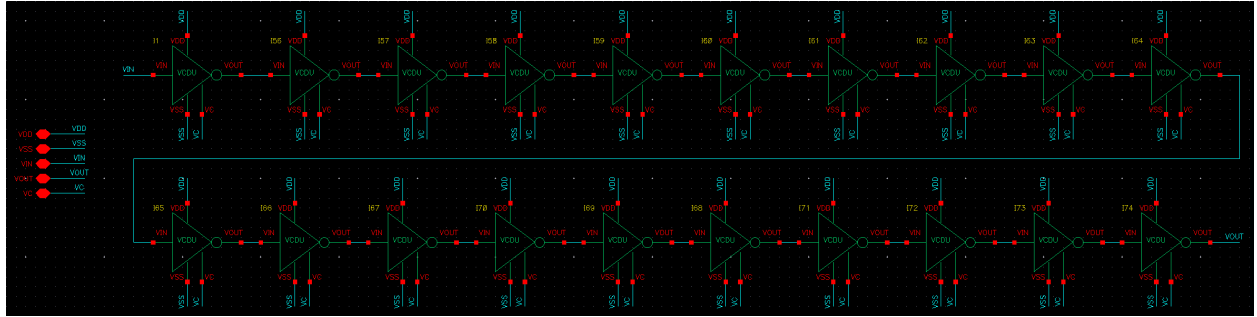


Figure 2: Complete VCDL architecture comprising 20 cascaded VCDU stages. The devices are sized such that the intrinsic input gate capacitance ($C_{ggN} + C_{ggP}$) of the subsequent stage serves as the primary load capacitance for the preceding unit.

2. VCDL Delay Characteristics (t_{pLH} vs V_c)

Figure 3 confirms the VCDL delay decreases monotonically across the 400 mV to 1.4 V tuning range. This ensures stable feedback and provides sufficient delay variation of 2.0 ns to 2.22 ns to cover the required 450 MHz to 500 MHz target frequencies across all PVT corners.

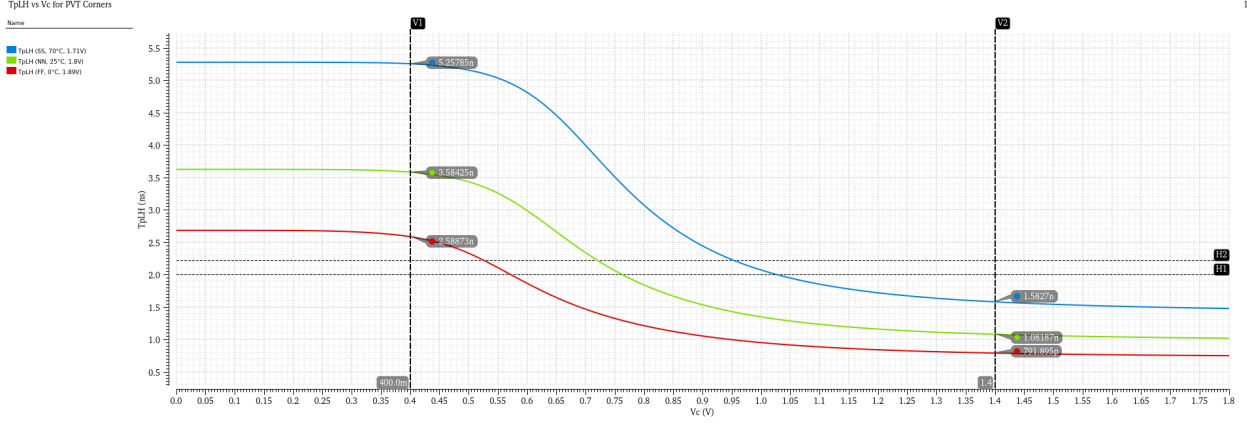


Figure 3: Total propagation delay (t_{pLH}) of the VCDL as a function of control voltage (V_c) evaluated across Fast (FF, 0 °C, 1.89V), Typical (NN, 25 °C, 1.8V), and Slow (SS, 70 °C, 1.71V) PVT corners for 50fF output capacitor load.

3. VCDL Gain (K_{vcdl} vs V_c)

Figure 4 illustrates the VCDL gain (K_{vcdl}), derived directly from the delay curve's slope. Extracting these K_{vcdl} values at the specific locking voltages is necessary to calculate the small-signal closed-loop bandwidth (f_{bw}).

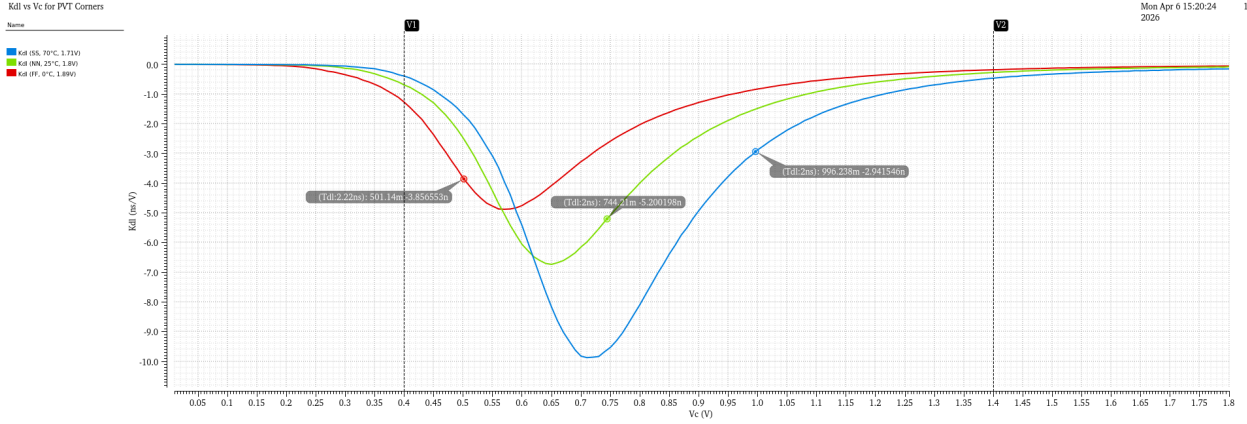


Figure 4: Voltage-Controlled Delay Line gain (K_{vcdl}) extracted as the derivative of the delay curve across all three PVT corners.

4. Power Consumption of VCDL (P_{dl} vs V_c)

Figure 5 demonstrates that power dissipation increases proportionally with control voltage, as delay elements draw more current to run faster. As expected, the Fast corner (FF) exhibits the worst-case power consumption due to its elevated 1.89 V supply.

7. Transient Locking Behavior of DLL Under 'Fast' Condition

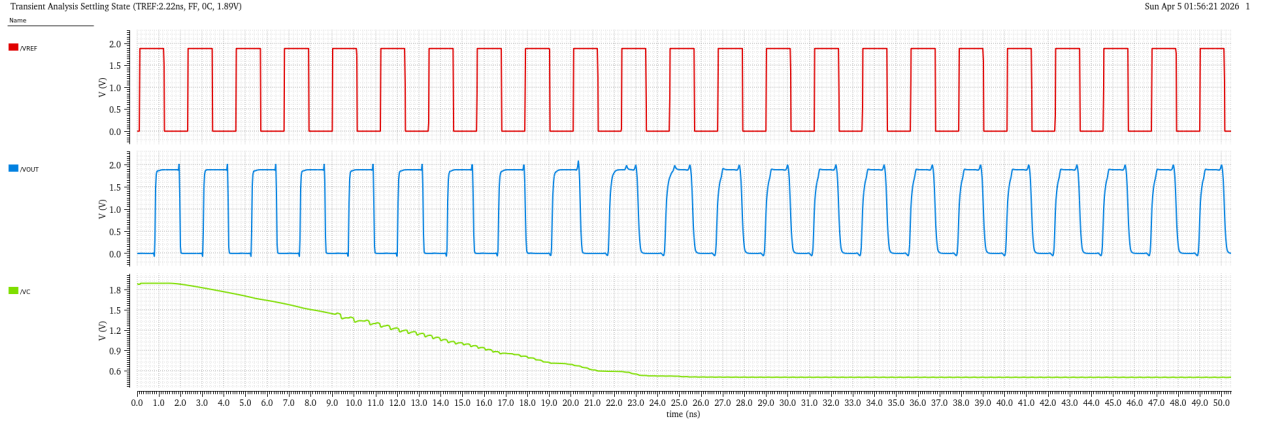


Figure 9: Transient startup and settling behavior of the DLL in the Fast (FF) corner (0°C, 1.89 V) tracking a 2.22 ns reference clock.

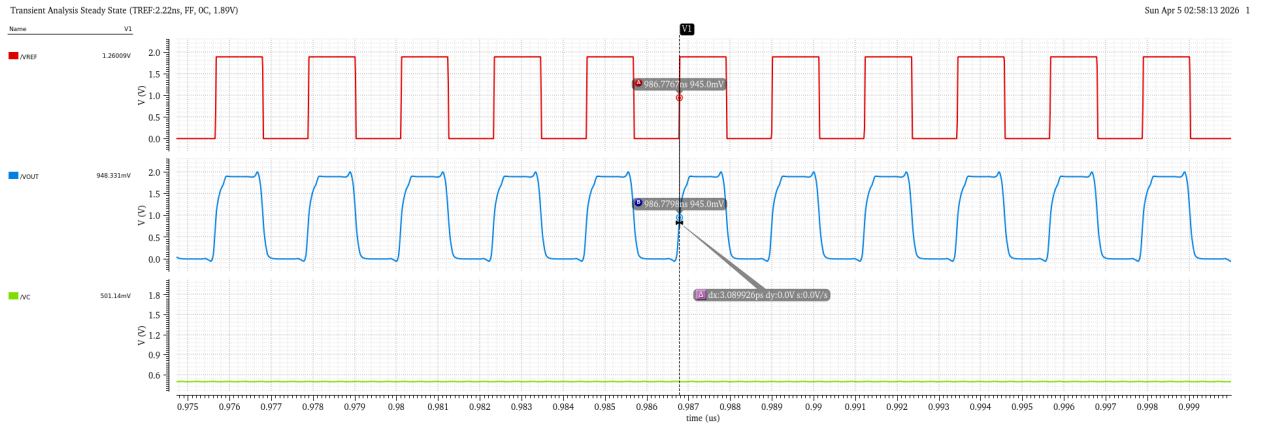


Figure 10: Steady-state locked condition in the Fast (FF) corner, demonstrating a Static Phase Offset (SPO) of approximately 3.08 ps.

Figure 9 illustrates the critically damped settling of the control voltage (V_C) to its locked state in the Fast corner, confirming loop stability with the 5 pF loop filter. Figure 10 verifies the steady-state performance, showing phase Locking with a Static Phase Offset (SPO) of 3.08 ps.

8. Transient Locking Behavior of DLL Under 'Slow' Condition

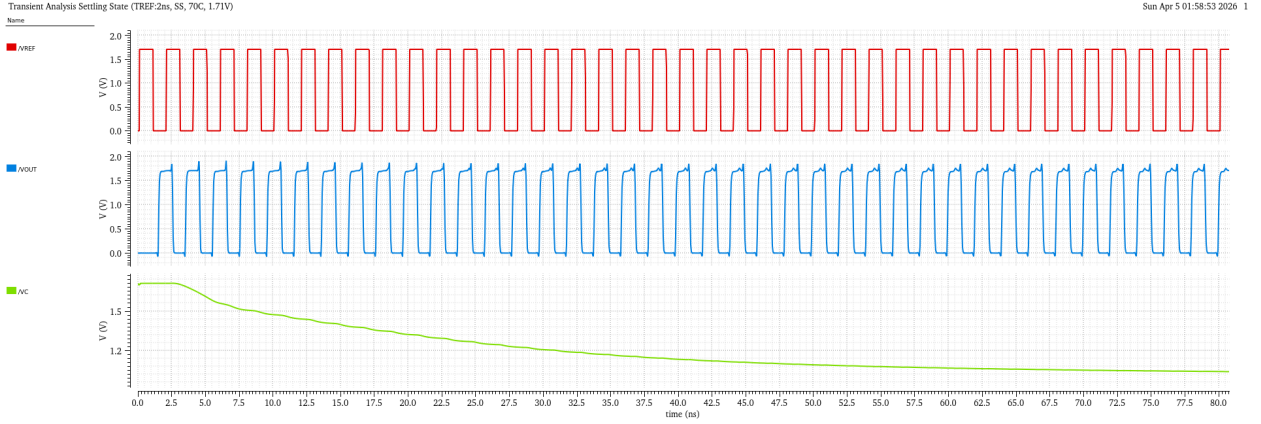


Figure 11: Transient startup and settling behavior of the DLL in the Slow (SS) corner (70 °C, 1.71 V) tracking a 2.0 ns reference clock.

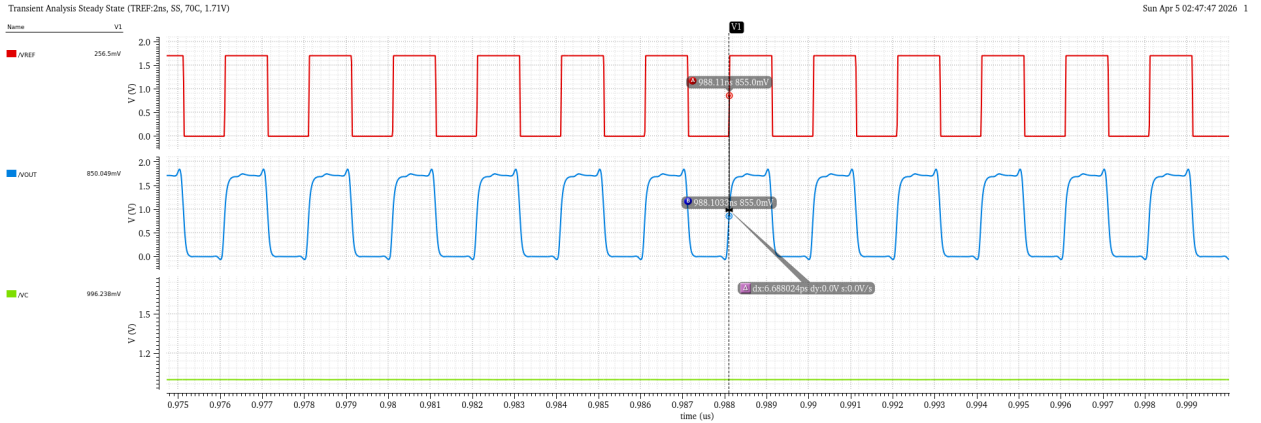


Figure 12: Steady-state locked condition in the Slow (SS) corner, demonstrating a Static Phase Offset (SPO) of approximately 6.68 ps.

Figure 11 confirms the Slow corner without any unstable ringing or harmonic locking. Figure 12 demonstrates that even under worst-case slow conditions, the system precisely locks with a final SPO of 6.68 ps, within acceptable design margins for the target frequency.

9. DLL Loop Bandwidth Calculation (Table 1)

The closed-loop tracking bandwidth (f_{bw}) of a first-order charge-pump DLL is calculated using the continuous-time linear approximation:

$$f_{bw} = \frac{I_{cp} \cdot K_{vcdl} \cdot f_{ref}}{2\pi \cdot C_{LF}} \quad (1)$$

where $I_{cp} = 20 \mu\text{A}$ is the charge pump current, and $C_{LF} = 5 \text{ pF}$ is the loop filter capacitor. The VCDL gain (K_{vcdl}) is extracted directly from the slope of the delay curve (Figure 4) at the final locked control voltage (V_{c0}) for each respective PVT corner.

9.1. Calculation for Fast (FF) Corner

For the FF corner tracking a $T_{ref} = 2.22 \text{ ns}$ ($f_{ref} \approx 450.45 \text{ MHz}$) clock, the loop locks at $V_{c0} \approx 0.501 \text{ V}$. The extracted VCDL gain at this operating point is $K_{vcdl} \approx 3.86 \text{ ns V}^{-1}$.

$$f_{bw(FF)} = \frac{(20 \times 10^{-6}) \cdot (3.86 \times 10^{-9}) \cdot (450.45 \times 10^6)}{2\pi \cdot (5 \times 10^{-12})} \approx 1.107 \text{ MHz}$$

9.2. Calculation for Slow (SS) Corner

For the SS corner tracking a $T_{ref} = 2.0 \text{ ns}$ ($f_{ref} = 500 \text{ MHz}$) clock, the loop locks at $V_{c0} \approx 0.996 \text{ V}$. The extracted VCDL gain at this operating point is $K_{vcdl} \approx 2.94 \text{ ns V}^{-1}$.

$$f_{bw(SS)} = \frac{(20 \times 10^{-6}) \cdot (2.94 \times 10^{-9}) \cdot (500 \times 10^6)}{2\pi \cdot (5 \times 10^{-12})} \approx 0.936 \text{ MHz}$$

PVT Corner	V_{c0} (Locked)	K_{vcdl} (Extracted)	Theoretical Bandwidth (f_{bw})
Fast (FF, 0°C , 1.89V)	0.501 V	3.86 ns V^{-1}	1.107 MHz
Slow (SS, 70°C , 1.71V)	0.996 V	2.94 ns V^{-1}	0.936 MHz

Table 1: Calculated closed-loop DLL bandwidth for Fast and Slow conditions.

Design Note: By actively sizing the loop filter capacitor up to 5 pF , the theoretical loop bandwidth was intentionally restricted to below 1.2 MHz . Because $f_{bw} \ll f_{ref}/10$, the discrete-time sampling effects of the PFD are largely suppressed, confirming that the transient plots exhibit a critically damped, ringing-free voltage Settling.

10. Static Phase Offset (SPO) Verification (Table 2)

The Static Phase Offset (SPO) measures the steady-state phase error between V_{REF} and V_{OUT} . Evaluated post-lock across PVT corners (Table 2). The worst-case SPO occurs in the Slow (SS) corner at 6.69 ps (within 25 ps specification limit) for the 2.0 ns clock period.

PVT Corner	Target Period (T_{ref})	Measured SPO
Fast (FF, 0 °C, 1.89 V)	2.22 ns	3.09 ps
Slow (SS, 70 °C, 1.71 V)	2.00 ns	6.69 ps

Table 2: Static Phase Offset (SPO) measured at steady-state locked conditions across PVT corners.

11. Steady-State Power Consumption (Table 3)

To isolate steady-state tracking power from dynamic settling power, the DLL’s power consumption was evaluated post-lock over an 800 ns to 1 μ s window. The average power was extracted in Cadence ADE using:

$$P_{avg} = \text{average}(\text{clip}(V_{DD} \cdot |I_{VDD}|, 800\text{n}, 1\text{u})) \quad (2)$$

PVT Corner	Supply (V_{dd})	Average Steady-State Power
Fast (FF, 0 °C)	1.89 V	5.009 mW
Slow (SS, 70 °C)	1.71 V	13.675 mW

Table 3: DLL steady-state power consumption across PVT corners.

Design Insight: Counter-intuitive that the Slow (SS) corner exhibits the highest power consumption. This is due to the characteristic of the VCDL architecture. In the Fast corner, the high intrinsic device mobility requires a very low locked control voltage ($V_{c0} \approx 0.526$ V) to achieve the required delay, resulting in low current draw. Conversely, in the Slow corner, the loop must drive V_c significantly higher ($V_{c0} \approx 1.028$ V) to force the slow delay elements to meet the 2.0 ns target, drastically increasing the steady-state current and resulting in a greater power of 13.675 mW relative to the Fast corner.

12. Design Methodology and Theoretical Analysis

12.1. Design Methodology

The DLL was optimized at the transistor level to ensure strict phase alignment and transient stability for a 450 MHz – 500 MHz reference clock (2.22 ns - 2.0 ns period).

- **VCDL Sizing Strategy:** The 20-stage Current-Starved Inverter VCDL eliminates explicit passive capacitors. Instead, devices are sized so the intrinsic input gate capacitance ($C_{ggN} + C_{ggP}$) of subsequent stages serves as the primary load.

- **TSPC PFD Optimization:** Initial designs utilized a standard Master-Slave D-Flip-Flop, but its excessive internal propagation delay (≈ 400 ps) bottlenecked high-frequency phase detection. Thus architecture was upgraded to a True Single-Phase Clock (TSPC) topology. To resolve the "ghost pulse" reset errors inherent to dynamic logic, a custom memory-wipe NMOS was integrated to asynchronously discharge internal nodes. Additionally, asymmetric stack sizing (wide data transistors, narrow clocked transistors) ensures fast D -to- Q propagation while minimizing reference clock loading.
- **Source-Switched Charge Pump Design:** It was designed to deliver a precise $I_{cp} = 20 \mu\text{A}$ using g_m/I_D methodology. To ensure high accuracy current mirroring, higher lengths were chosen ($L_n = 2.0 \mu\text{m}$, $L_p = 1.12 \mu\text{m}$) maximize output resistance, with widths sized for $V_{OV} \approx 150 \text{ mV}$ via I_D/W curves. An op-amp was added to actively track V_c at internal nodes, suppressing channel-length modulation to perfectly match UP/DOWN currents across a wide dynamic range.
- **Loop Dynamics:** To eliminate underdamped V_c ringing observed during initial simulations ($I_{cp} = 20 \mu\text{A}$), the loop filter capacitor was increased from initial 1.13 pF to 5 pF . This increased the damping factor (ζ), ensuring critically damped, smooth exponential settling across all PVT corners.

12.2. Explanation of Divergence from Theory

While simulation closely tracks first-order linear models, expected deviations arise from large-signal analog behaviors:

- **Bandwidth Approximation Limits:** The physical DLL's discrete-time sampling nature and large-signal behavior difference alter actual settling time versus small-signal f_{bw} predictions. Specifically, initializing V_c near V_{DD} severely reduces the charge pump PMOS V_{DS} , temporarily forcing it into the triode region. This drops I_{cp} below the nominal $20 \mu\text{A}$ design value, changing initial settling compared to ideal continuous-time models.
- **Dynamic Non-Idealities and Static Phase Offset (SPO):** Ideal phase-locking assumes zero steady-state error, yet the system exhibits a minor SPO (up to 6.69 ps in the SS corner, Table 2). While the op-amp-regulated source-switched charge pump, highly eliminates static UP/DN mismatches by actively matching V_{DS} and shields the loop filter from severe charge injection and clock feedthrough, dynamic non-idealities remain. The residual SPO is primarily driven by internal node charge-sharing (dynamic charging/discharging of source parasitics) and non-negligible practically implemented PFD's turn-on and off asymmetries. To maintain a net-zero charge on the loop filter Capacitor at steady state, the system naturally enforces this small phase error to compensate for these non-ideal effects.